

3.3 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions.

a) $y = x^8 - 3\sqrt{x} + \frac{5}{x^3}$

$$y = x^8 - 3x^{1/2} + 5x^{-3}$$

$$\frac{dy}{dx} = 8x^7 - \frac{3}{2}x^{-1/2} - 15x^{-4}$$

b) $q = 5x^e + \sin(e^\pi) - \ln(5) + 2e^x$

$$\frac{dq}{dx} = 5ex^{e-1} + 0 - 0 + 2e^x = 5ex^{e-1} + 2e^x$$

(Note that $\sin(e^\pi)$ and $\ln(5)$ are both constants and have a derivative equal to 0)

c) $y = e^x \left(\frac{1}{2}x^{10} - 2x \right)$

Use Product Rule

$$\frac{dy}{dx} = e^x (5x^9 - 2) + e^x \left(\frac{1}{2}x^{10} - 2x \right) = e^x \left[\frac{1}{2}x^{10} + 5x^9 - 2x - 2 \right]$$

d) $w = x \left(\frac{x}{\sqrt{x^3}} \right)^4$

Simplify first!!!

$$w = x \left(\frac{x}{x^{3/2}} \right)^4 = x \left(\frac{x^4}{x^6} \right) = \frac{1}{x} = x^{-1}$$

$$\frac{dw}{dx} = -1x^{-2} = \frac{-1}{x^2}$$

e) $y = \frac{2t}{5-t^5}$

Use Quotient Rule

$$\frac{dy}{dt} = \frac{(5-t^5)(2) - (-2t)(-5t^4)}{(5-t^5)^2} = \frac{2(5-t^5) + 10t^5}{(5-t^5)^2} = \frac{10+8t^5}{(5-t^5)^2}$$

f) $T = \frac{\sqrt[3]{k} + k^{-2}}{k}$

Simplify first!!!

$$T = \frac{\sqrt[3]{k} + k^{-2}}{k} = \frac{\sqrt[3]{k}}{k} + \frac{k^{-2}}{k} = k^{-2/3} + k^{-3}$$

$$\frac{dT}{dk} = -\frac{2}{3}k^{-5/3} - 3k^{-4}$$

2. Suppose that $f(5)=1$, $f'(5)=6$, $g(5)=-3$, & $g'(5)=2$. Find each of the following:

a) $(fg)'(5)$

$$(fg)'(5) = f'(5)g(5) + f(5)g'(5) = (6)(-3) + (1)(2) = -18 + 2 = -16$$

b) $\left(\frac{f}{g}\right)'(5)$

$$\left(\frac{f}{g}\right)'(5) = \frac{g(5)f'(5) - g'(5)f(5)}{[g(5)]^2} = \frac{(-3)(6) - (2)(1)}{(-3)^2} = \frac{-20}{9}$$

c) $\left(\frac{g}{f}\right)'(5)$

$$\left(\frac{g}{f}\right)'(5) = \frac{f(5)g'(5) - f'(5)g(5)}{[f(5)]^2} = \frac{(1)(2) - (6)(-3)}{(1)^2} = 20$$

d) $\left(\frac{fg}{g-f}\right)'(5)$

$$\begin{aligned} \left(\frac{fg}{g-f}\right)'(5) &= \frac{[g(5)-f(5)](fg)'(5) - [g(5)-f(5)]'[f(5)g(5)]}{[g(5)-f(5)]^2} \\ &= \frac{[g(5)-f(5)][f(5)g'(5) + f'(5)g(5)] - [g'(5)-f'(5)][f(5)g(5)]}{[g(5)-f(5)]^2} \\ &= \frac{[-3-1][(1)(2) + (6)(-3)] - [2-6][(1)(-3)]}{[-3-1]^2} = \frac{[-4][-16] - [-4][-3]}{(-4)^2} = \frac{64-12}{16} = \frac{52}{16} = \frac{13}{4} \end{aligned}$$

3. What is the equation of line tangent to the graph of $y = 2\sqrt{x^3} + 3x^2 + x + 7$ at its y -intercept?

The y -intercept of the graph occurs when $x = 0$, so $y(0) = 2\sqrt{(0)^3} + 3(0)^2 + (0) + 7 = 7$.

So the y -intercept occurs at the point $(0, 7)$.

We rewrite the given function: $y = 2\sqrt{x^3} + 3x^2 + x + 7 = 2x^{3/2} + 3x^2 + x + 7$

We then calculate the derivative: $y' = 3\sqrt{x} + 6x + 1$.

So the slope of the tangent line is given by: $y'(0) = 3\sqrt{0} + 6(0) + 1 = 1$

Therefore, the equation of the tangent line is given by:

$$y - 7 = 1(x - 0)$$

$$y = x + 7$$

4. If $f(2) = 10$ and $f'(x) = x^2 f(x)$ for all x , find $f''(2)$.

$$f''(x) = x^2 f'(x) + 2xf(x)$$

$$f''(2) = (2)^2 f'(2) + 2(2)f(2) = 4f'(2) + 4(10) = 4[(2)^2 f(2)] + 40 = 4(40) + 40 = 200$$

5. Find the x -value for each point on the graph of $f(x) = \frac{5}{4}x^4 - \frac{14}{3}x^3 - 3x^2$ where the line tangent to the curve is parallel to the line $y = 0$.

$$f'(x) = 5x^3 - 14x^2 - 6x = x(5x^2 - 14x - 6)$$

Using the Quadratic Formula we can factor $5x^2 - 14x - 6$.

$$x = \frac{14 \pm \sqrt{196 - 4(5)(-6)}}{10} = \frac{7 \pm \sqrt{79}}{5}$$

$$\text{Therefore we have: } f'(x) = x \left(x - \frac{7 + \sqrt{79}}{5} \right) \left(x - \frac{7 - \sqrt{79}}{5} \right)$$

We want $f'(x) = 0$. This happens when:

$$x = 0, \frac{7 + \sqrt{79}}{5}, \text{ \& } \frac{7 - \sqrt{79}}{5}$$

6. Find the equations of the tangent lines to the curve $y = \frac{x^2}{x+1}$ that are parallel to the line $6x + 2y = 1$.

There derivative of our function is $y' = \frac{(x+1)(2x) - (x^2)(1)}{(x+1)^2} = \frac{2x^2 + 2x - x^2}{(x+1)^2} = \frac{x^2 + 2x}{(x+1)^2}$.

We need to find a tangent line that is parallel to the line $6x + 2y = 1 \Leftrightarrow y = \frac{1}{2} - 3x$ which has a slope of -3. Therefore, we want an x -value that will make the derivative equal to -3.

$$\begin{aligned} -3 &= \frac{x^2 + 2x}{(x+1)^2} \rightarrow -3(x+1)^2 = x^2 + 2x \rightarrow -3x^2 - 6x - 3 = x^2 + 2x \rightarrow 4x^2 + 8x + 3 = 0 \\ &\rightarrow (2x+1)(2x+3) = 0 \rightarrow x = -\frac{1}{2} \text{ or } x = -\frac{3}{2} \end{aligned}$$

These x -values correspond to the points: $\left(-\frac{1}{2}, \frac{1}{2}\right)$ & $\left(-\frac{3}{2}, -\frac{9}{2}\right)$.

Therefore, the equations of our lines are:

$$y - \frac{1}{2} = -3\left(x + \frac{1}{2}\right) \Leftrightarrow y = -3x - 1$$

$$y + \frac{9}{2} = -3\left(x + \frac{3}{2}\right) \Leftrightarrow y = -3x - 9$$

7. At what points, if any, does the graph of $y = x^3 - 3x + 4$ have a horizontal tangent?

$$y' = 3x^2 - 3$$

The slope of a horizontal tangent is 0. So we want to know what values of x make $y' = 0$.

$$0 = 3x^2 - 3 \rightarrow x^2 = 1 \rightarrow x = \pm 1$$

A horizontal tangent occurs at the points: $(-1, 6)$ & $(1, 2)$.

8. Find equations of the tangent line and the normal line the curve $y = x\sqrt{x}$ at the point $(1,1)$.

We rewrite the given function as $y = x\sqrt{x} = x \cdot x^{1/2} = x^{3/2}$.

We should first check that the problem makes sense. We check to see if the point $(1,1)$ is on the graph.

$f(1) = (1)^{3/2} = 1$. So the point $(1,1)$ is on the graph.

We then calculate the derivative of the function: $f'(x) = \frac{3}{2}x^{1/2}$.

The slope of the tangent line at $(1,1)$ is then given by: $m = f'(1) = \frac{3}{2}(1)^{1/2} = \frac{3}{2}$.

Equation of the tangent line through $(1,1)$:

$$y - 1 = \frac{3}{2}(x - 1)$$

$$y - 1 = \frac{3}{2}x - \frac{3}{2}$$

$$y = \frac{3}{2}x - \frac{1}{2}$$

Equation of the normal line through $(1,1)$:

The normal line must have the opposite reciprocal slope: $m = -\frac{2}{3}$

$$y - 1 = -\frac{2}{3}(x - 1)$$

$$y = -\frac{2}{3}x + \frac{5}{3}$$

9. Compute $f''(x)|_{x=1}$ if $f(x) = x^2(x-2)$.

This means that we must evaluate $f''(x)$ when $x = 1$.

$$f(x) = x^2(x-2) = x^3 - 2x^2$$

$$f'(x) = 3x^2 - 4x$$

$$f''(x) = 6x - 4$$

$$f''(1) = 6(1) - 4 = 2$$

10. Consider the function $y = 5x^{-1} - \frac{1}{5}x$.

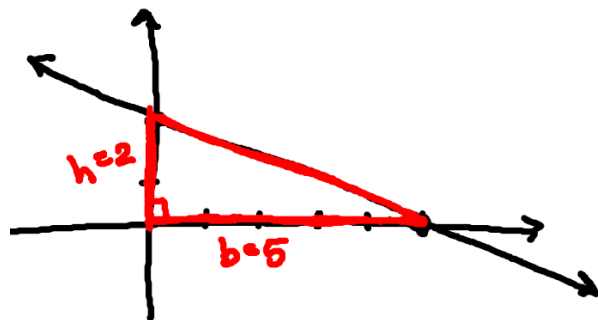
a) Find the equation of the tangent line to the curve at the point $(5, 0)$.

We compute the derivative: $y' = -5x^{-2} - \frac{1}{5}$. We know $m = y'(5) = -5(5)^{-2} - \frac{1}{5} = -\frac{5}{25} - \frac{1}{5} = -\frac{2}{5}$.

Therefore the equation of the tangent line is: $y - 0 = -\frac{2}{5}(x - 5) \Leftrightarrow y = -\frac{2}{5}x + 2$

b) Find the area of the triangle formed from the coordinate axes and the tangent line to the curve at the point.

We can then sketch the triangle and see that: $b = 5$ & $h = 2$.



So the area of the triangle is:

$$A = \frac{1}{2}bh = \frac{1}{2}(5)(2) = 5$$

TRUE/FALSE

1. If f and g are differentiable, then $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

TRUE: This is the Sum Rule

2. If f and g are differentiable, then $\frac{d}{dx}[f(x)g(x)] = f'(x)g'(x)$

FALSE: We must use the Product Rule

3. If $y = e^2$, then $y' = 2e$

FALSE: e^2 is a constant and its derivative is 0.

4. The derivative of a polynomial is a polynomial.

TRUE: Using the Power Rule or Constant Rule on terms of a polynomial will always result in a polynomial.